Class 10 NCERT Maths
CLASS 10 NCERT MATHS
Chapter 11: Areas Related to Circles
CHAPTER 11: AREAS RELATED TO CIRCLES

A simplified, detailed guide in Hinglish
For students who missed Class 7-8-9
Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

1.1 CIRCLE KE BASIC FORMULAS (YAAD RAKH)

Circumference (perimeter) = 2 x pi x r   (or pi x d)
Area of circle = pi x r^2
Diameter = 2r

r = radius
pi = 22/7 (or 3.14)

1.2 pi (PI) KYA HAI?
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pi = circumference / diameter ka ratio.
Har circle ke liye SAME hota hai = 3.14159...

EXAM MEIN:
- Radius 7 ka multiple ho   -> use pi = 22/7
- Warna   -> use pi = 3.14

1.3 EXAMPLE: BASIC

EXAMPLE 1: Radius 7 cm circle ka area aur circumference

Area = pi r^2 = (22/7) x 7 x 7 = 22 x 7 = 154 cm^2
Circumference = 2 pi r = 2 x (22/7) x 7 = 44 cm

SECTION 2: SECTOR OF A CIRCLE

2.1 SECTOR KYA HAI?

SECTOR = circle ka ek "pizza slice" - do radii aur
ek arc ke beech ka area.

Angle theta us slice ka angle hai (centre pe).

```
       arc
      _____
    /       \
   / sector \
  /  theta   \
 *-----O------*
 radius  radius
```

2.2 FORMULAS (theta in degrees)

Area of sector   = (theta/360) x pi x r^2
Length of arc   = (theta/360) x 2 x pi x r
Perimeter of sector = arc length + 2r

LOGIC: Pura circle 360 degree hai. Sector us ka
theta/360 hissa hai. So area aur arc bhi utna fraction.

2.3 EXAMPLES

EXAMPLE 2: Radius 21 cm, angle 60. Sector area.

Area = (60/360) x (22/7) x 21 x 21
     = (1/6) x (22/7) x 441
     = (1/6) x 22 x 63
     = (1/6) x 1386
     = 231 cm^2

EXAMPLE 3: Radius 7 cm, angle 90. Arc length.

Arc = (90/360) x 2 x (22/7) x 7
    = (1/4) x 44
    = 11 cm

SECTION 3: SEGMENT OF A CIRCLE

3.1 SEGMENT KYA HAI?

SEGMENT = chord aur arc ke beech ka area (sector mein se triangle nikaal do).

```
       ____
      /  seg  \
     /.________.\    <- chord
    /  triangle \
   *------O------*
```

3.2 FORMULA

Area of (minor) segment
= Area of sector - Area of triangle

Area of sector = (theta/360) x pi x r^2
Area of triangle = (1/2) x r^2 x sin(theta)

So:
Segment = (theta/360) pi r^2 - (1/2) r^2 sin(theta)

3.3 EXAMPLE

EXAMPLE 4: Radius 10 cm, angle 90. Minor segment area.
(pi = 3.14)

Sector area = (90/360) x 3.14 x 10 x 10
= (1/4) x 314 = 78.5 cm^2

Triangle area = (1/2) x r^2 x sin 90
= (1/2) x 100 x 1 = 50 cm^2

Segment = 78.5 - 50 = 28.5 cm^2

SECTION 4: MINOR VS MAJOR

- MINOR sector/segment: chhota wala (angle < 180)
- MAJOR sector/segment: bada wala (angle > 180)

- Major area = total circle area - minor area
- Major angle = 360 - minor angle

Q AND A TIME

Q1. Radius 14 cm circle ka area aur circumference.
(pi = 22/7)

Q2. Radius 6 cm, angle 60. Sector area. (pi = 3.14)

Q3. Radius 14 cm, angle 90. Arc length. (pi = 22/7)

Q4. Sector ka perimeter kaise nikaalte hain?
(formula bata)

Q5. Radius 4 cm, angle 90. Minor segment area.
(pi = 3.14, sin 90 = 1)

Q6. Minor aur major sector mein kya difference hai?

SUMMARY

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1. Circumference = 2 pi r, Area = pi r^2

2. pi = 22/7 (radius 7 ka multiple) ya 3.14

3. Sector area = (theta/360) x pi r^2

4. Arc length = (theta/360) x 2 pi r

5. Perimeter of sector = arc length + 2r

6. Segment area = sector area - triangle area
   = (theta/360) pi r^2 - (1/2) r^2 sin(theta)

7. Major = total - minor

TIPS FOR EXAM

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1. theta/360 ka fraction pehle simplify kar lo
   (jaise 90/360 = 1/4) - calculation easy.

2. pi = 22/7 vs 3.14 sahi choose karo - radius dekho.

3. Segment = sector - triangle. Triangle ke liye
   (1/2) r^2 sin(theta) formula yaad rakh.

4. Units: area = cm^2, length/perimeter = cm.

5. Combination figures (square mein circle, etc.) mein areas ko ADD/SUBTRACT karo - diagram banao.